HW 9 – Requirements Doc

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Version and Change Log

V0.1 – Stakeholders and user goals	10/29/26
V0.2 – Added purpose and scope section	10/30/26
V0.3 – Added functional and non-functional reqs	10/30/26
V0.3.5 – Fixed consistency between req IDs and traceability matrix	10/30/26
V0.4 – Add Moscow Summary	10/30/26
V0.5 – Add assumptions Section	10/30/26
V1.0 – Add inherited constraints	10/30/26
V1.1 – Added a few extra could and won't sections to MoSCoW	10/30/26

Purpose and Scope

SkillForge's main aim is to improve the usability and capabilities of autonomous robot learning from exo-centric human demonstrations. The system will be used to convert human demonstration videos into robot policy models that can perform the demonstrated task in simulation. SkillForge provides a workflow for robotics engineers to select videos, configure training parameters, review what they have selected, start training, and monitor training progress.

In scope actions include:

- Selecting and uploading human demonstration videos

- Validating video file types and sizes
- Configuring training parameters
- Reviewing selected videos and training configurations
- Sending training requests to the backend
- Monitoring training status and available progress information
- Provide access to generated model artifacts
- Displaying understandable validation and processing errors
- Supporting project constraints

Stakeholders and User Stories

Stakeholder Map:

Primary	Secondary	Hidden
Robotics Engineer - Automatic extraction from demonstration.	**VP investing in robotics development** – Company investment leader	**Robot Teleoperator** – Spend less time manually curating data.
Robotics Engineer - Training process optimization for camera quality and viewpoint specifications.	**ML-ops Team Lead -** Requirement of standard formats.	

US-01: As a **VP of a manufacturing company**, I want to **develop advanced robotics automation capabilities** so that **my company minimizes costs and stay competitive.**

US-02: As a **Robotics Engineer,** I want **the system to extract relevant actions automatically from a demonstration,** so **I do not have to manually label each step.**

US-03: As a **ML-ops team lead,** I want **the final model to be available in standard formats,** so that my **team does not spend time rewriting our proprietary backend to support the model format.**

US-04: As a **Robot Teleoperator,** I want the **robot to learn from easily generated exo-centric data**, so I **can spend less time curating manual robot demonstration data.**

US-05: As a **Robotics Engineer,** I want **the training process to require minimal requirements for camera quality and viewpoint specifications,** such that **data collection and training can fit my use-case easily.**

Functional Requirements

ID	Statement	MoSCoW	Source Story	Verification Method
REQ-01	The system shall display and provide a demonstration video.	Must	US-02	Verify that a demonstration video is present on the website.
REQ-02	The system shall provide automatic validation for video file type and size.	Must	US-03	Video file type and size is analyzed on upload and a verification message is shown to the user displaying a summary of the video and any error messages if it is denied.
REQ-03	The system shall provide users manual configuration for any training/hyper parameters.	Must	US-05	Website has prompts for all training/hyper parameters, alongside a short description for each when hovered.
REQ-04	The system shall provide a review of the selected video and configuration before processing.	Must	US-03	Verify that a review is displayed to the user when they finish setting the configuration.
REQ-05	The system shall send a training request to the backend with training parameters	Must	US-02	Verify with logs generated by the backend that the training request was processed

REQ-06	The system shall provide a training status window to the user.	Should	US-01	Verify that a training status window is displayed when training request is accepted
REQ-07	The system shall display the generated model artifact reference.	Should	US-04	GUI should display the generated model artifact reference when training process is finished.
REQ-08	The system shall display error messages at any point in the training process with an error code that provides an explanation.	Should	US-01	Purposefully enter invalid arguments/corrupted data to test errors. GUI should display error code and backend should generate a log.
REQ-09	The system shall provide camera-quality and view-point guidance.	Could	US-05	Guidance is given before training process begins(?).
REQ-10	The system shall provide a documented model-artifact format for downstream management.	Could	US-04	Model-artifact information is provided in the summary generated after training.

Non-functional Requirements

ID	**Statement**	**MoSCoW**	**Source Story**	**Verification Method**
NFR-01	The system shall have an API response time of 300ms (this is just an estimation)	Should	US-01	During the 10 test runs, debugging tools can be used to measure response time.
NFR-02	The system shall require authentication	Must	US-01	Test run of 10 operations of verified and non-

	for any protected training-run operations.			verified users interacting with all protected operations and verifying that non-verified users are denied.
NFR-03	The system shall have a usability rate of 80% positive from a sample of the intended user base.	Should	US-02	Usability category from sampled reviews is examined to determine the positive rate.
NFR-04	The system shall preserve in-progress training status in the event of a restart on the backend.	Should	US-03	5 test runs of trained models where the process is interrupted at different critical points in the training process; system successfully resumes after restart
NFR-05	The system shall clean up all uploaded training data for a model after its use.	Must	US-03	5 Trained models with verification that all training data has been deleted in-between each training session.

MoSCoW Summary

Must Have:

- REQ-01 – Select and display a demonstration video
- REQ-02 – Validate video file type and size
- REQ-03 – Configure and validate training parameters
- REQ-04 – Review the selected video and configuration

- REQ-05 – Submit a training request and display its identifier and initial status
- NFR-02 – Require authentication for protected training-run operations
- NFR-05 – Delete uploaded training data after use

Should Have:

- REQ-06 – Display training-run status
- REQ-07 – Display the generated model artifact reference
- REQ-08 – Display error messages
- NFR-01 – Meet the proposed API response-time target
- NFR-03 – Meet the proposed usability target
- NFR-04 – Preserve persisted training-run status across backend restarts

Could Have:

- REQ-09 – Provide camera-quality and viewpoint guidance
- REQ-10 – Provide a documented model-artifact format for downstream management

Won't Have:

- Direct deployment of trained policies onto physical robots
- Guaranteed successful policy generation from every uploaded demonstration
- Guaranteed development of advanced robotics research capabilities beyond the team's expertise

Economic Constraints:

- CON-01 – Video storage must use subsections under 25 GB, shared through student OneDrive accounts
- CON-02 – Neural network training and inference must operate within a maximum GPU memory limit of 24 GB VRAM
- CON-03 – Average GPU computing costs must not exceed $0.17 per hour, based on the RTX 3090 pricing estimate
- CON-04 – The project should take advantage of available Modal AI credits, up to $150 per month

Professional Constraints:

- CON-05 – The project scope must reflect the team’s current expertise in robotics and neural-network research, prioritizing the product website, frontend, backend, and infrastructure
- CON-06 – The team should seek guidance from a project advisor and relevant professors to address gaps in robotics and machine-learning expertise

Security Constraints:

- CON-07 – Future customer payment processing must securely handle payment information
- CON-08 – Payment information must not be stored or retained after the payment session
- CON-09 – Training data must be fully deleted after it has been used
- CON-10 – The system architecture should balance low-cost computing with appropriate data security and allow flexibility for future deployment
- CON-11 – The deployment approach should support portability, such as through Docker containers

Inherited Constraints

The following constraints have been inherited from our team's earlier design-constraints work and retained under the IDs used in this document. Each constraint limits the implementation or determines the design choice.

CON-01 — Video storage must use subsections under 25 GB shared through student OneDrive accounts. Design implication: Video datasets must be partitioned and storage allocation managed within the stated limit.

CON-02 — Neural-network training and inference must operate within 24 GB of GPU VRAM. Design implication: Models, batch sizes, and processing stages must fit the available GPU memory; larger configurations require adjustment.

CON-03 — Average GPU compute cost must not exceed $0.17 per hour, based on the RTX 3090 estimate. Design implication: Compute selection and training duration must be planned against this cost ceiling.

CON-04 — The project should use available Modal AI credits, up to $150 per month. Design implication: Hosted processing should prioritize available credits and monitor consumption.

CON-05 — Scope must reflect the team’s current robotics and neural-network expertise, prioritizing the website, frontend, backend, and infrastructure. Design implication: The initial release cannot depend on inventing new robotics research methods.

CON-06 — The team should seek advisor and professor guidance on robotics and ML knowledge gaps. Design implication: Unresolved technical decisions require expert consultation rather than unsupported assumptions.

CON-07 — Future customer payment processing must handle payment information securely. Design implication: If payments enter scope, the system must use a secure payment flow; payment handling is not part of the current D0–D2 design.

CON-08 — Payment information must not be stored or retained after the payment session. Design implication: Any future payment integration must avoid persistent storage of payment details.

CON-09 — Training data must be fully deleted after use. Design implication: The storage lifecycle must include deletion of uploaded training data after its intended use; retention and deletion timing require team confirmation.

CON-10 — The architecture should balance low-cost computing with data security and future deployment flexibility. Design implication: Hosting and storage choices must account for cost, protected access, and portability rather than optimizing cost alone.

CON-11 — Deployment should support portability, such as Docker containers. Design implication: Services should avoid unnecessary host-specific dependencies and remain containerizable.

Assumptions and Open Questions:

Assumptions

*N/A Date means there is no real solution – the assumption will always exist as a constraint.

Assumption	Owner	Date
Input video is either multi-view (3+ views), or has available depth sensor data	Noah Schumahcer	N/A
Input video is of a human performing an action – 1 human per video	Noah Schumahcer	N/A
Video contains enough data to train model to replicate demonstrated capability	Ryan Broyles	N/A

Open Questions

Question	Owner	Date
What are the key components that will need to be optimized to reduce computing power?	Ryan Broyles	12/02/2026
How will the system handle deformable items in videos?	Noah Schumacher	01/14/2027

Glossary

API – Application Programming Interface; A set of rules and protocols allowing different software applications to communicate and share data with another.

GPU – Graphics Processing Unit; hardware used to accelerate neural-network training and inference.

VRAM – Video Random Access Memory; memory available on GPU for storing model and training data.

Model Artifact – A saved output of training that can be stored, retrieved, or used by another system.

Modal AI – A cloud computing platform used by the team to access compute resources and available credits.

Robotics Engineer – The primary user responsible for preparing demonstrations and configuring robot training.

Robot Teleoperator – A person who remotely operates a robot and may provide demonstration data.

VP – Vice President; a company leader who is responsible for business decisions.

Simulation – A virtual environment used to test or train robot behavior without requiring a physical robot.

Robot Policy – A learned model or control strategy that determines how a robot should act in a given situation.

Demonstration Video – A recording of a human performing a task that a robot is intended to learn.

Exo-centric – A third-person viewpoint from a camera that records an activity from outside the person who is performing it.

Training Run – An execution of the configured model training process.

Training Configuration – The collection of parameters used to control a training run.

Training Data – Demonstration videos and data used during model training.

Inference – Using a training model to generate predictions or actions.

Neural Network – A machine learning model composed of interconnected computational units that learns patterns from data.

Advisor Review

Meeting Date: 5:30 PM 10/30/26

Document Suggestions:

1: Glance over user stories and see if they are still specific to the use cases of the application

2. Consider adding more to the Could and Won't list in the MoSCoW summary